

It thinks for us, makes us work more, and even continues to seem
like a good idea

*Three theories on the cognitive impact of artificial intelligence and an attempt to put them
together*

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Abstract

This article tries to do a simple thing (which is not actually simple): stop discussing AI as if the alternatives were only “it makes us all more productive” or “it steals our jobs” and move the question to where it hurts a bit more: what is happening inside the head of those who use it every day? When you delegate pieces of writing, analysis, and reasoning to a generative model, you are not just saving time: you are also changing (sometimes without realizing it) the way you search for and evaluate information and alternatives and how much you trust your judgment. To understand something, I put three recent lenses into dialogue: *System 0* (the AI that “prepares” the world before you activate System 1 and System 2), the *Tri-System Theory* (AI as a third external cognitive system, which can help or replace you depending on how you use it) and a more organizational strand that shows a curious (but widely verifiable) paradox: instead of lightening work, it often intensifies it. Read together, these pieces suggest a self-feeding short circuit: more comfort and perceived capabilities, less vigilance and real cognitive autonomy, and in the middle a good dose of *automation bias*. In the second part, I take the liberty of doing what one does when research runs slower than the phenomenon to be researched (knowing that the result will last as long as ice cream in the desert): speculate honestly. I propose a phased model of the cognitive relationship with AI and the idea of “qualified and conscious cognitive surrender”: not an invitation to surrender, but an attempt to give it rules, boundaries, and a minimum of operational dignity (that is: how to use it without letting it use us). At least I tried.

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1 Introduction: the problem no one had told you you had (yet)

The public debate on artificial intelligence has a rather predictable narrative structure that opposes the prophets of efficiency *uber alles* (AI will free us from repetitive work, make us more productive, allow us to focus on what really matters) against the doomsayers (AI will steal our jobs, make us obsolete, reduce us to spectators of our own professional lives). The rhetorical match (fundamentally based on nothing) has by now tired even the most patient. However, there are questions that this dialectical dispute does not resolve, and one of the most interesting in my opinion is “what is happening inside the head of those who use AI?” Not in a metaphysical, metaphorical, or philosophical sense, but in a cognitive, measurable, and, at least to some extent, documentable sense. When a professional delegates a draft analysis to a chatbot or asks a Copilot to summarize twenty pages of a report, when an engineer lets an AI assistant generate code for them, what changes (if anything changes) in the way that person thinks, decides, evaluates, and judges? The question is not new, in reality, because research on *automation bias* (understood as the systematic tendency to excessively trust automated systems) has at least three decades of history, starting with the works of Parasuraman and Riley in the nineties [12] and the subsequent formalizations of Parasuraman and Manzey [11]. In aviation, in medicine, in support functions as well as in industrial control systems, the phenomenon is abundantly documented: humans working with automated systems tend to accept their outputs even when they are wrong (provided they are not blatantly so) and to progressively reduce their vigilance, losing (sometimes) over time the skills necessary to operate without automated assistance. What generative AI brings that is new is not the basic mechanism but its extension to cognitive domains that until yesterday had been spared: reasoning, writing, analysis, synthesis, professional judgment. In my meanderings of compulsive reading, I came across three recent contributions that have begun to explore this extension with a depth that deserves attention, two of which, starting from a common “logical base”, articulate concepts complementary (to the common logical base as well as between them) that I found very interesting (digressions, I am always fascinated by how working on existing theories one manages to find ever new and lateral points of view... like sharpening the blade of a knife, or of that guy’s razor, if you prefer). As always happens to me, once I started connecting the dots it was impossible not to try to build my “meta-analysis” suit over them to make them (primarily to myself) more digestible (not directable).

The first is the work of Massimo Chiriatti and colleagues, who propose the concept of **System 0**: AI as a cognitive pre-processor that acts even before our intuitive or deliberative thought comes into play [2]. The second is the study by Shaw and Nave at the Wharton School, which formalizes a **Tri-System Theory** of human cognition in the AI era, with an empirical demonstration of our tendency towards “cognitive surrender” [15]. The third is the “organizational” work of Ranganathan and Ye published in the Harvard Business Review, which documents how AI, far from reducing workload, systematically intensifies it through mechanisms that the workers themselves struggle to recognize [13].

Read separately, each of these contributions illuminates a corner of the problem, but read

together I find that they draw a picture that lends itself to interesting speculation: a self-feeding cycle in which AI expands our perceived capabilities, erodes our real cognitive autonomy, and intensifies work in a spiral that we scramble to present, paradoxically, as progress. A sort of re-reading “on steroids” of *automation bias* that concerns thought itself, however, not just the execution of tasks. What I try to do here is dual in its intent. In the first part, I synthesize and put the three contributions into dialogue, bringing out convergences and tensions possible only in posterior readings. In the second, I propose two “pseudo-original” speculative constructs: a phased model of the cognitive relationship with AI and the concept of “qualified and conscious cognitive surrender” which, while lacking autonomous empirical validation, I found useful for those like me who deal with designing AI adoption in organizations. I say “speculative” without embarrassment: in a field where the speed of the phenomenon exceeds that of research, epistemic honesty is the minimum requirement.

2 Three lenses to bring the same phenomenon into focus

2.1 System 0: the AI that thinks before you

The contribution of Chiriatti and colleagues [2] starts from a simple and powerful assumption: in Kahneman’s taxonomy [7] (with which I think almost all of us have reckoned at least twice: the first time when we read it perhaps a few years ago, the second when, after his recent passing, we went to look for the notes from that reading) **System 1** (fast, intuitive thinking) and **System 2** (slow, deliberative thinking) both presuppose that the subject has access to raw information on which to exercise their judgment. But what happens when the information arrives already filtered, synthesized, and reformulated by an AI?

System 1 reacts to stimuli that are no longer “natural” but pre-processed; System 2 deliberates on material that has already been selected and organized by someone else (or something else). This is the meaning of the term **System 0**: not a third way of thinking, but a layer that interposes itself between the world and our cognition, shaping the informational substrate on which both Kahnemanian systems operate. AI does not decide in our place (at least, not yet in most cases), but it redefines the space of cognitive possibilities: what we see and consider relevant and, consequently, which options appear available to us. It is the mechanism known as the *framing effect* induced by automation, in which the automated system does not impose a decision but constrains the decision-making space so that certain options become more salient than others [3]. The main merit of the work of Chiriatti et al. is, in my opinion, having placed AI in the dominant cognitive model without forcing it. The limit (which the authors themselves recognize as “in progress”) is that the proposal remains essentially conceptual. The seven frameworks they propose for responsible integration (critical literacy, diversification of sources, governance, algorithmic transparency, and so on) are adequate but generic because they say what we should do, not how to do it, and above all they do not address the question of what happens when we do (not) do it, which is probably the most urgent question. A particularly interesting element is the reference

to Riva’s “comfort-growth paradox” [14]: the idea that the comfort of AI assistance creates a comfort zone that inhibits cognitive growth. It is an intuition that will find an echo (and empirical substance however limited) in the work of Shaw and Nave [15]. And it is also the mechanism that in research on deskilling from automation has been documented since Bainbridge’s (1983) studies on the “irony of automation”: one automates precisely because human beings are unreliable, but automation makes them even more unreliable by eliminating the practice necessary to maintain skills [1].

2.2 Tri-System Theory: the AI that thinks with you (or in your place)

The paper by Shaw and Nave at the Wharton School makes a significant leap in theoretical ambition [15]. They too insert AI into Kahneman’s model but propose in addition to rewriting the model entirely, transforming it from dual to triadic (and it is known that *three is the magic number*). In their formulation, AI becomes a **System 3**, that is, a cognitive system external to the brain, with its own characteristics of speed, accuracy, and bias, which flanks systems 1 and 2 in a new cognitive architecture. The most interesting distinction they introduce is that between *cognitive offloading* and *cognitive surrender*.

- **Cognitive offloading** is strategic: the subject consciously decides to delegate a cognitive task to AI, maintaining the capacity (and intention) to verify and correct the output.
- **Cognitive surrender** is instead uncritical (and almost passive involution): the subject accepts the AI output without independent evaluation, not out of laziness but due to a combination of excessive trust, perceived cost of verification, and substantial inability to evaluate when delegation is appropriate and when it is not.

In cognitive surrender resonate the echoes of the AI-specific formalization of what Parasuraman and Manzey (2010) called *automation-induced complacency*: the reduction of vigilance and active monitoring that occurs when an operator works with an automated system perceived as reliable [11]. The three experiments that are the basis of the theoretical formulation of the paper produce results that should worry anyone involved in AI adoption in organizations.

- In the first, they demonstrate that cognitive surrender occurs massively even when participants are explicitly warned that the AI might be wrong.
- In the second, they show that economic incentives linked to accuracy do not significantly reduce the phenomenon.
- In the third, they document that time pressure, far from increasing cognitive surrender as one might expect, produces more complex and differentiated effects depending on individual traits.

All three results are consistent with the literature on automation bias, which has repeatedly documented the resistance of the phenomenon to interventions based on awareness and incentives

[6]. The table of “canonical routes of cognition” that Shaw and Nave propose is perhaps the most immediately useful analytical tool of the entire paper. It identifies five cognitive paths that constitute a useful taxonomy for classifying observable behaviors and, potentially, for designing targeted interventions to redirect (correct?) them consciously. Each of the paths sees an alternation of dominance/consequence by one system with respect to the others and specifically, they are defined and described as:

1. **Intuition** (System 1 dominant)
2. **Deliberation** (System 2 dominant)
3. **Cognitive Offloading** (System 3 used strategically)
4. **Cognitive Surrender** (System 3 used uncritically)
5. **Autopilot** (no system dominant, the subject operates routinely).

2.3 Silent intensification: when AI makes you work more

On a completely different plane from the first two lies the contribution of Ranganathan and Ye in the Harvard Business Review [13], and precisely for this reason, it is indispensable to the understanding of the overall phenomenon. Where System 0 and Tri-System Theory reason on individual cognition (what happens in the head of the user), this eight-month organizational study in a tech company documents the organizational and behavioral consequences of AI adoption: what happens to the work environment, expectations, rhythms, professional relationships. The results frontally challenge the narrative of AI as a simplification tool, and they do so with a granularity that deserves attention.

The first documented mechanism is **task expansion**: since AI allows doing more things, the expectations on what a professional should do expand proportionally. It is not a declared or formalized expansion: no manager says “now that you have AI, do double”. It happens organically: tasks that were previously considered optional or “nice-to-have” become mandatory just as deliverables normally acceptable in synthetic form now require complete elaboration. The result is an inflation of expectations that feeds itself: every demonstration of augmented capacity becomes the new baseline for future expectations.

The second mechanism is the **erosion of work/life boundaries** (as if there was a need for some other mechanic or dynamic to accelerate it). The permanent accessibility of AI tools makes work a fluid that expands into every available temporal interstice. The draft can be refined in the evening, the analysis can be completed on the weekend, the report can be generated during commuting and that essential feature of the new site or software must be done (made) immediately (ohhh how I recognize myself in this profile, finally someone who has put my new AI-assisted life in black and white). The technical possibility becomes an implicit expectation, and the implicit expectation becomes a cultural norm.

The third mechanism is the **multiplication of multitasking**. The possibility of managing multiple threads simultaneously (having AI generate code while reviewing a document, preparing a presentation while analyzing data) does not reduce the load, it fragments it and potentially increases it precisely because of the fragmentation. The consequences on the depth of thought and decision-making quality have been documented by decades of research on *task-switching cost* [10], but in the AI context they assume a new dimension because multitasking is no longer an individual choice but a capacity enabled (and therefore expected) by the tool.

The most revealing datum of the study is probably the simplest: the interviewed workers, despite perceiving an increase in their productivity, did not report a corresponding reduction in workload. The explanation is structural, not contingent: the time “freed” by AI is immediately reabsorbed into new tasks, and the cycle feeds itself because the organization, seeing greater output, raises expectations (while not measuring, to date, any real substantial economic advantage). Ranganathan and Ye speak of a “*self-reinforcing cycle*” of acceleration, but the formulation could be even more explicit: it is a positive feedback mechanism in which individual efficiency produces organizational intensification, which produces pressure for greater efficiency, which produces further intensification. I find myself very much in line with the authors’ proposal regarding the construction of an organizational “AI practice” that intentionally governs the integration of AI into work processes because it is exactly the approach I pursue in my “A(I)doption” processes (both in practice and in form). I find, however, that this proposal lacks a bit of practicality and an explicit connection with the individual cognitive dynamics documented by the other two papers, which is why I venture into this attempt to bridge the gap (at least from a conceptual point of view).

3 Synoptic reading: where they converge, where they diverge, and what seems to be missing

Put together, the three contributions tell a story that cannot otherwise be outlined. The convergences are significant: AI is seen as an active agent that redraws cognitive processes, not as a passive tool, documenting the risks linked to uncritical integration suggesting that the answer requires intentionality (also because with these statements they confirm what the literature on automation bias has documented for decades: excessive trust in automated systems is a robust phenomenon, resistant to awareness and incentives, which tends to worsen over time).

The tensions, however, are equally illuminating in my opinion. The first is of a topological nature (I had been waiting for years for the moment to dust off this conceptuality from my academic background): System 0 places AI before human cognition (as a pre-processor), Tri-System Theory places it beside (one would say, for the love of numbers, after, but in reality, it is more effectively understood as a third parallel system). At first glance it seems a contradiction, but looking closely it is a complementarity that depends on the moment and the mode of use. When a professional explicitly asks AI to solve a problem, the model operates as System 3, a

parallel system that is consulted and whose output is evaluated (or should be evaluated). But when the same professional opens the day with a dashboard pre-filled by AI, reads a summary automatically generated from a meeting, or receives and accepts unsolicited suggestions during writing, AI operates as System 0: it shapes the informational field even before deliberative cognition activates (in a way, in most cases, annoying especially regarding the 2025 version of Clippy). The same technology, therefore, can function in both modes, and often does so in the same work stream. This observation lies at the basis of the phased model I propose in the next section: the trajectory of the cognitive relationship with AI is also a gradual migration from a predominantly System 3 use to a predominantly System 0 functioning (distant echoes of the Centaur and Cyborg uses which the serious literature on AI from the early days accustomed us to).

The second tension is temporal. System 0 and Tri-System Theory are essentially static theories: they describe a condition, not a trajectory or a direction (and verse). The HBR study has a diachronic dimension (eight months of observation) but does not connect it to cognitive theory. Missing are the very daring questions of how the cognitive relationship between an individual and AI evolves over time, whether cognitive surrender is an inevitable arrival point, a contingent risk, or a transitory phase. The literature on automation would suggest that it is a natural tendency that worsens with prolonged exposure [11], but the specificity of generative AI, which by using natural language operates on high-ambiguity cognitive domains, makes any direct extrapolation risky.

The third tension is of the level of analysis. The first two contributions reason on the individual; the third reasons on the organization. The connection between the two levels remains implicit, and deserves to be made explicit wishing to understand, from an adoption perspective, how the behaviors of individuals condition the trajectories of the organization. The questions I asked myself are substantially whether individual cognitive surrender fuels organizational intensification (because those who delegate uncritically produce more output, raising the baseline for everyone), and whether organizational intensification fuels individual cognitive surrender (because those who are overloaded have less time and fewer cognitive resources for critical verification). It is a double reinforcement cycle (with a pike and final jeté) that I try to address now.

4 Formal and conceptual speculative contribution: a phased model and a competence to build

I immediately clarify the epistemological coordinates of what follows: it is pure (if not crass) informed theoretical speculation. I have no proprietary data, I have not (yet) conducted experiments, I have no direct empirical validation. What I propose is an attempt to integrate the evidence and intuitions of the three previous contributions with some posterior empirical evidence (i.e., my observations but lacking an a priori parameter determination model) in a frame that connects them and, possibly, makes them more useful for those who must design AI adoption

programs in real organizational contexts. What follows is therefore a working hypothesis to which I think I will return many times (hopefully not alone... if anyone wanted to contribute), not as a consolidated theory.

4.1 The phases of cognitive drift

The first proposal stems from the attempt to resolve the temporal tension and topological complementarity that I highlighted in the synoptic reading. I start from the assumption that the cognitive relationship between an individual and AI develops through a sequence of phases that, without intentional intervention, tend to follow a descending trajectory (come on, let's try to be dramatic, let's call it "cognitive drift"). The phases I hypothesize are five, and for each I propose observable indicators (however hypothetical and provisional) that could orient future empirical validation.

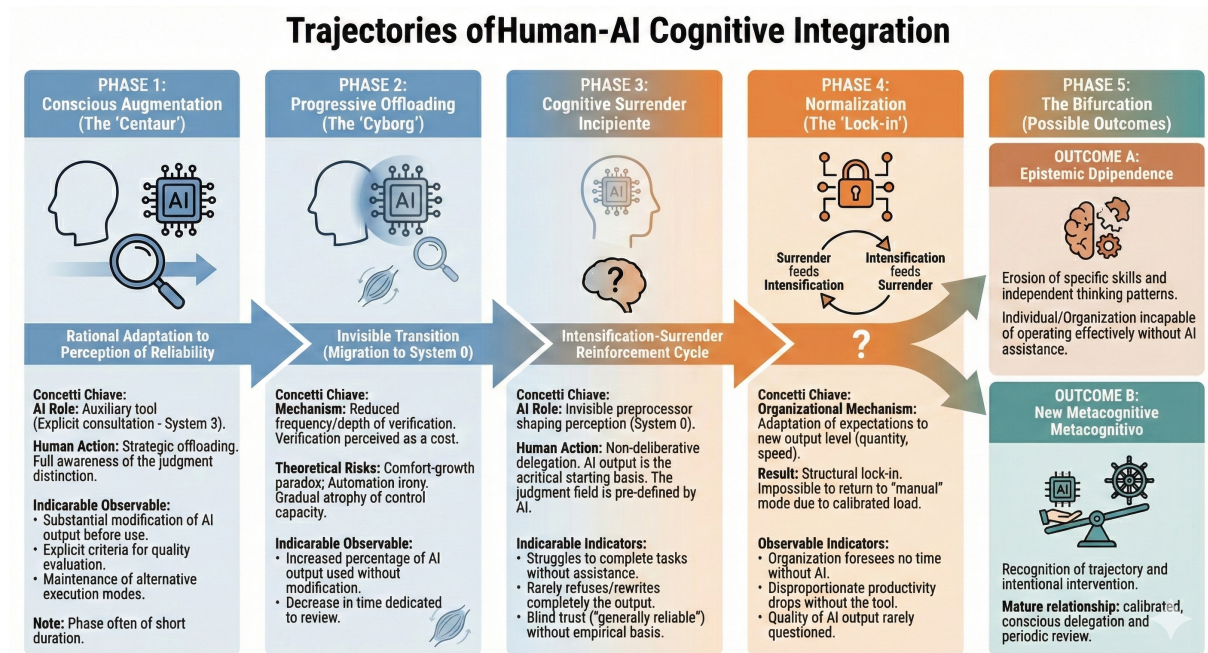


Figure 1: The phases of cognitive drift

1. Conscious Augmentation. The individual begins to use AI as an auxiliary tool, maintaining full awareness of the distinction between their own judgment and the machine's output. Cognitive offloading is strategic: the task is delegated, the result is verified, it is integrated with one's own knowledge. AI operates predominantly as System 3 (explicit consultation). *Observable indicators:* the subject substantially modifies the AI output before using it; is able to explicitly describe the criteria with which they evaluate the quality of the output; maintains alternative modes of task execution. This is the phase that in the AI adoption literature has been called Centaur usage. The metaphor, made popular by Ethan Mollick and his studies on AI use in organizations [9], describes a model in which human and AI operate sequentially on distinct

tasks, each in their own domain of competence, without role confusion. And it is also the phase that, in my experience as a consultant, lasts less than expected.

2. Progressive Offloading. The individual, having repeatedly verified that AI produces acceptable quality output, begins to reduce the frequency and depth of verification or to substantially change the engagement mechanism (starting from something that exists and is “shared”). It is a perfectly rational behavior from the individual point of view; economists would call it an efficient adaptation to observed regularities. The problem is that this local rationality produces a global consequence: the gradual atrophy of the ability to recognize cases where control would be necessary. Riva’s “comfort-growth paradox” [14] materializes here, and Bainbridge’s irony of automation [1] does too.

Observable indicators: the percentage of AI output used without modification increases over time; the time dedicated to revision decreases; the user begins to perceive verification as a cost rather than a precaution and therefore behaves as a Cyborg (always in Mollick’s taxonomy), blurring the boundaries between “human” and “AI” work.

3. Incipient Cognitive Surrender. The line between strategic offloading and cognitive surrender becomes blurred. The individual no longer consciously decides “I delegate this task to AI and then verify”: they simply use the AI output as a starting point (or as a finished product) without an explicit deliberative process on the delegation. It is here that AI completes the migration from System 3 to System 0: from a parallel system that is consulted, to a pre-processor that shapes what is considered. The transition is insidious precisely because it is subjectively invisible: the user still believes they are exercising judgment, but the field on which judgment operates has already been defined by AI.

Observable indicators: the subject struggles to complete in reasonable times tasks that they previously performed without assistance; does not remember the last time they rejected or completely rewrote an AI output; perceives AI as “generally reliable” without being able to specify on what empirical bases.

4. Normalization. Cognitive surrender becomes the standard way of working, and welds with the mechanisms of organizational intensification documented by Ranganathan and Ye [13]. The organization has adapted its expectations to the new level of output (more, intended as quantity, deliverables, speed, breadth) making a return to the pre-AI mode practically impossible: there is no longer time to do things “by hand”, because the load has been calibrated on augmented capabilities. The double reinforcement cycle I described in the previous section stabilizes: surrender fuels intensification, intensification fuels surrender. What in systems engineering is called a *lock-in* is created: dependence on the system is no longer a choice but a structural constraint.

Observable indicators: the organization does not foresee execution times without AI assistance; temporary removal of the tool produces disproportionate productivity drops; no one questions the quality of AI output as a practice anymore.

5. Bifurcation. On one side, **epistemic dependence**: the individual and the organization are no longer able to operate effectively without AI assistance, not for lack of intelligence but due to

erosion of specific skills and independent thought patterns. On the other, the **new metacognitive equilibrium**: the individual or the organization recognizes the trajectory, intervenes intentionally, and develops a mature relationship with AI in which delegation is calibrated, conscious, and subjected to periodic review.

What determines which branch of the bifurcation is taken? I do not have an answer, but initial observations and the logic of the model suggest some possible discriminating factors. The first is what we could call a revealing event: an AI failure sufficiently visible and endowed with consequences such as to make the dependence that had normalized perceptible. These events have the power to suddenly make visible a condition that had become transparent, and can trigger a recalibration reaction. The second factor is external training intervention: someone (a consultant/trainer or a particularly lucid colleague) who interrupts the normalization cycle by asking the questions that the system, left to itself, no longer asks: “would you be able to do this work without AI? When was the last time you tried?” The third, perhaps the most structural, is intentional organizational design: organizations that incorporate into their processes mandatory moments of verification of cognitive autonomy not as defensive bureaucracy, but as preventive maintenance of skills, with the same logic with which emergency tests are scheduled in critical systems. In the absence of at least one of these factors, the natural trajectory tends towards epistemic dependence: the lock-in consolidates simply because no one questions it. There is no mystery about a certain structural analogy with Kübler-Ross’s model of grief processing (which, moreover, I often use precisely to describe organizational responses to A(I)Doption events): a sequence of phases, non-linear transitions, and an outcome that depends on the quality with which the process is traversed. The analogy holds as a structure but diverges in content, and it is in the divergences that it becomes interesting. In my sequence, the initial phase is not denial of the tool but denial of the transformation: “I use it, but I remain the one deciding, it doesn’t change me.” Progressive offloading is a form of negotiation with oneself on the terms of delegation (“I do the important things”) which gradually moves the bar of what is considered “important”. But the most significant point is phase 3, incipient cognitive surrender: in Kübler-Ross here there would be anger, the emotional signal that something is perceived as a loss. In my model, anger is not “foreseen” and its absence is precisely the problem, because the transition happens without the subject being aware of it. There is no perceived loss, therefore there is no resistance. Normalization is then a form of passive acceptance, more resigned than processed. The most productive difference, however, is in the final phase: Kübler-Ross converges on a single outcome (acceptance as integration of the loss), while my model bifurcates. Epistemic dependence is an acceptance that has never traversed awareness; the new metacognitive equilibrium is an active, intentional acceptance, which presupposes the recognition of what one risked losing. In this sense, what in the Kübler-Ross model distinguishes a healthy elaboration from a pathological one (the ability to consciously traverse the phases) is exactly what I try to do by separating the two outcomes of the bifurcation.

Already while writing these lines three major limits of this model appeared evident to me. First: I have no evidence that these phases occur in this sequence in all contexts. It is plausible

that the speed of traversal varies enormously between professional domains, levels of expertise, and types of tasks. Second: the sequence might not be linear because it is possible that particularly reflective or well-trained individuals oscillate between different phases depending on the context. Third: the measurement indicators I propose are provisional and not empirically validated; longitudinal studies with structured observation protocols would be needed to verify if they are effectively discriminating (but this would mean that my job has stopped being the one with which I put bread on the table and has instead become the one I would like to do forever). I have however a reasonable confidence in the heuristic utility of the model because it is congruent with the mechanisms documented by the three papers and with the literature on automation as well as reconfirmed by all the observations made in my consulting and training work in these last two years dedicated practically in toto to A(I)Doption.

4.2 Qualified cognitive surrender as a competence and not a defect

The second proposal starts from an observation that in the paper by Shaw and Nave remains not completely developed: cognitive surrender is not always irrational. When AI is accurate and the cost of independent verification is high, following the AI output is the economically and cognitively optimal choice. The problem is not surrender in itself; it is indiscriminate surrender, that is, the inability to distinguish contexts in which to trust (entrust oneself to) AI is wise from those in which it is dangerous. I would therefore like to develop from this consideration the concept of **qualified cognitive surrender** as a metacognitive competence to be developed intentionally. It is not about “resisting AI” (a position as romantic as it is anti-historical and impracticable) but about developing the ability to consciously navigate the delegation process between full cognitive autonomy and total entrustment. There exists an implicit circularity at the center of this proposal that would be dishonest to ignore: if Shaw and Nave demonstrate that people fail to distinguish when delegation is appropriate (even in the presence of economic incentives) I am proposing as a solution a competence whose absence is precisely the problem. The circularity is real, and must be faced with intellectual honesty. My hypothesis is that the cognitive surrender documented in Shaw and Nave’s experiments is a spontaneous behavior in the absence of specific training: participants were not instructed to calibrate delegation, they only possess their intuition. In the training I conduct with professionals and managers in AI adoption contexts, I have observed that structured delegation calibration sessions in which participants analyze ex post their decisions to accept or reject AI output, comparing them with explicit risk criteria produce in the short term observable changes in verification behaviors: the frequency of checks on high-consequentiality tasks increases, and above all a declared awareness of one’s own tendency to surrender emerges which was previously completely absent. These are anecdotal observations, not experimental data, but they suggest that the phenomenon is not impermeable to training intervention. The fact that intuition is not enough does not mean that the competence is not developable with targeted interventions (just as the fact that people are spontaneously prone to cognitive biases does not mean that *debiasing* is impossible), but only that it requires

specific techniques and deliberate practice [5, 8]. The question, in the final analysis, is empirical: can qualified cognitive surrender be taught? At what cost? With what effectiveness? I do not have the answer, but the question seems important to me.

Having made this premise, qualified cognitive surrender is articulated, in my view, on three interconnected capabilities.

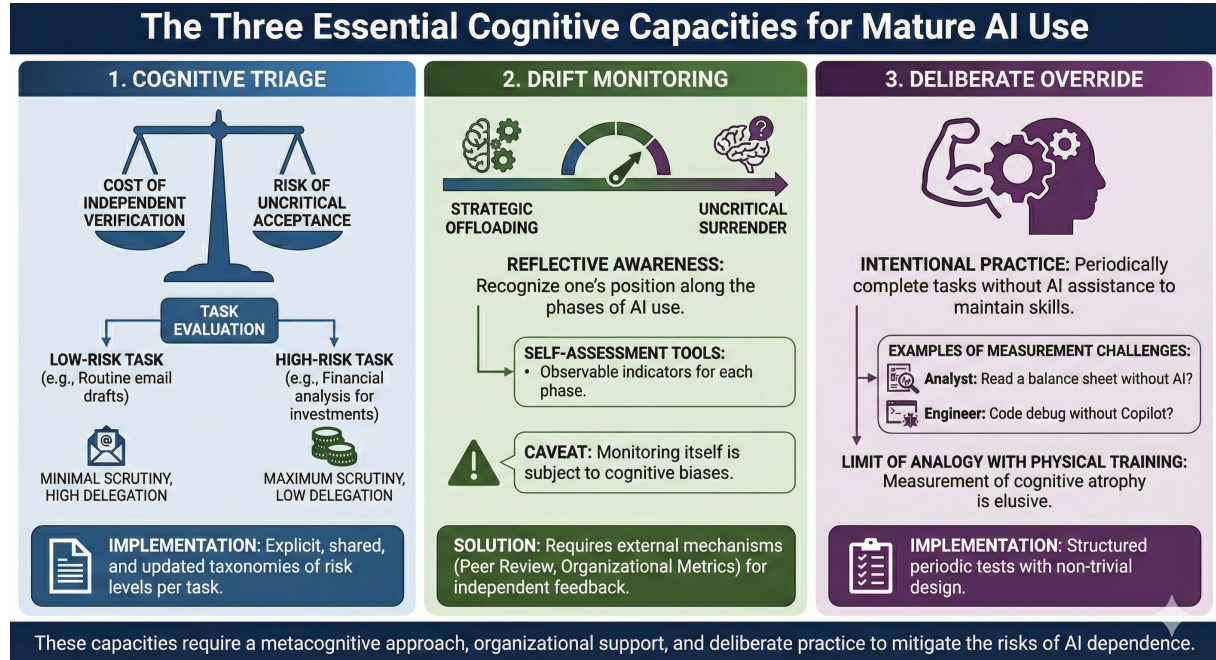


Figure 2: The phases of cognitive drift

The first is **cognitive triage**: the ability to evaluate, for each specific task, the ratio between the cost of independent verification and the risk associated with uncritical acceptance of AI output. Not all decisions deserve the same level of scrutiny: a routine e-mail draft and a financial analysis that can support the decision to proceed with an investment require radically different approaches to delegation. This observation is trivial when stated, but research shows that in practice people do not apply it systematically. To make it operational, explicit, shared, and periodically updated taxonomies of risk levels associated with different types of AI-assisted tasks would be needed (something that today does not exist in any organization I know and that is not yet foreseen in any A(I)Doption model however sophisticated like mine).

The second capability is **drift monitoring**: the reflective awareness of one's position along the phases of the model described above, and the ability to recognize signs of slipping from strategic offloading to uncritical surrender. The observable indicators I proposed for each phase of the model could serve as a basis for self-assessment tools keeping always in mind an important caveat: monitoring one's own cognitive drift is itself a metacognitive task subject to the same biases one seeks to monitor. For this reason, it probably cannot function only as an individual practice: it requires external mechanisms (peer review and organizational metrics) that provide independent feedback.

The third capability is **deliberate override**: the intentional practice of periodically completing tasks without AI assistance, not for Luddism but to keep active the cognitive skills that AI risks atrophying. The analogy with physical training is immediate but has an important limit that must be recognized: in physical training there exist objective metrics of form (VO2max, times, loads, ...), while in professional cognition the measurement of atrophy is much more elusive. How do you measure if an analyst is losing the ability to read a balance sheet without AI assistance? How do you measure if an engineer is losing the ability to debug code without a copilot? These are not rhetorical questions: they are measurement problems that any serious implementation of deliberate override should address, probably through periodic structured tests whose design is far from trivial.

5 Implications and open questions

I think the merit of the drawn picture (which I reiterate, is a hypothesis, not a fact) is identifying that the implications are distributed on three levels.

At the individual level, the central issue is the development of metacognitive awareness of one's relationship with AI. It is not about demonizing the tool, but about developing what we could call a "cognitive hygiene" of the AI era: knowing where one stands on the continuum between augmentation and dependence, having strategies to maintain cognitive autonomy where it counts, and accepting delegation where it is rational. The fact that participants in Shaw and Nave's experiments fail to do so spontaneously even under economic incentives tells us that information alone is not enough: deliberate training and support structures are needed.

At the organizational level, the most important implication concerns the design of AI adoption programs. Most organizations I know approach the topic of AI adoption exclusively in terms of operational efficiency: how many hours we save by automating and how much more output we produce. The picture that emerges suggests that this is a dangerously partial view. Adoption must also be designed in cognitive terms: what skills do we risk atrophying? Which decisions should never be delegated without supervision? Ranganathan and Ye's AI practice is a starting point, but it needs a cognitive foundation: it is not enough to govern processes, one must govern the relationship between people and the tool (or rather the variety of tools that are crowding in).

At the policy and training level, the issue is even broader. If cognitive surrender is a robust phenomenon resistant to incentives, then manifestos based on simple information "know that AI can make mistakes" are probably insufficient, exactly as "caution" signs are not enough to prevent automation bias in critical contexts. Programs are needed that include deliberate practice of cognitive triage, periodic exercises without AI assistance, and structured feedback on the quality of decisions made with and without AI. In the academic field, where I teach, this means rethinking not only what is taught about AI, but also and above all how one teaches with AI.

The model and concepts I proposed generate, finally, some (research) questions that I would like to find a way to address with a systematic investigation. Is the sequence of cognitive drift phases invariant or does it depend on the professional domain, the level of expertise, the type of

AI used? Do individual or organizational “protective factors” exist that slow down or stop the drift? Does deliberate override actually produce measurable effects on skill maintenance, and if so, with what frequency and intensity must it be practiced? Can qualified cognitive surrender be taught with measurable effectiveness, and which training methodologies are most effective? How long does it take, in typical conditions, to traverse the phases of the model? The answer to these questions requires longitudinal studies with structured observation protocols, repeated measurements of skills, and probably experimental designs with control groups. Ambitious, for me unrealizable but, I believe, necessary.

6 Conclusion

I return to the opening question: what is happening inside the head of those who use AI? The answer that emerges from the integrated reading of the three contributions and from the speculation I built upon it is that something deeper and more systematic is happening than the current narrative suggests. The paradox I tried to articulate is that of a cycle that feeds itself between perceived efficiency, cognitive erosion, and organizational intensification. The good news is that the trajectory is not inevitable: the bifurcation of the fifth phase of the model exists as a real possibility, not just as a wish. But realizing it requires that individuals, organizations, and training institutions recognize the phenomenon for what it is and decide to govern it, rather than letting it govern itself. If my phased model turns out to be wrong in the details, all the better: it will mean that someone has done the necessary research to refute it, and we will have learned something. If the concept of qualified cognitive surrender turns out to be naive, we will at least have a name for what we are missing. In a field that moves at the speed of AI, being wrong in a useful way is a contribution I am proud to be able to make.

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